



Generative and Agentic AI

When AI Stops Suggesting and Starts Doing



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Overarching Agenda

10:00 AM – 10:45 AM (MDT)

- **MATLAB for Research:** low-code data analysis, latest updates, interoperability

11:00 AM – 12:00 PM (MDT)

- **Agentic AI Workflows:** available tools, leveraging MATLAB, and accelerating research workflows

Slides & File access: <https://github.com/drLKeen>

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Your MathWorks Team



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Frequently Asked Questions

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Artificial Intelligence in the USGS Ecosystems Mission Area ACTIVE

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Frequently Asked Questions

OFFICE OF SCIENCE QUALITY AND INTEGRITY

What is generative AI how can it be used products intended

By [Office of Science Quality and Integrity](#)

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or other

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Artificial intelligence strategy for the U.S. Geological Survey

February 18, 2026

[View Report](#)

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[HTML Format](#)

[USGS AI Strategy](#)

[AGU AI Policies](#)

Two Very Different Results

How to go from this

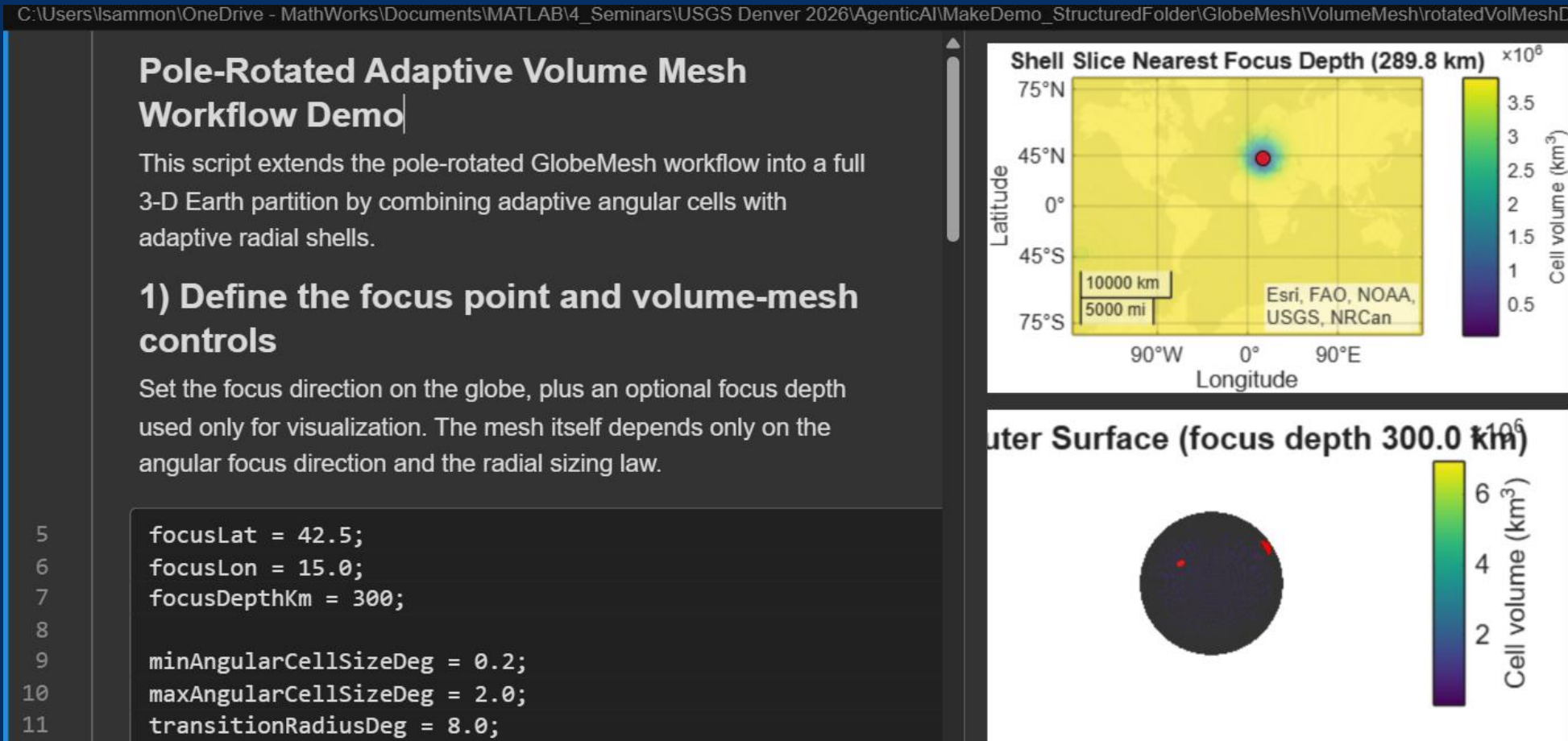
```
C:\Users\lsammon\OneDrive - MathWorks\Documents\MATLAB\4_Seminars\USGS Denver 2026\AgenticAI\MakeDemo_N...
9
10 %% Retrieve the latest NDBC observations
11 stationId = "COVM2";
12 dataUrl = "https://www.ndbc.noaa.gov/data/realtime2/" + sta
13
14 observations = readNdbcRealtime(dataUrl);
15 observations.Time = datetime(2000 + observations.YY, observ
16     observations.DD, observations.hh, observations.mm, 0, .
17     "TimeZone", "UTC");
18
19 % Focus the charts on the most recent seven days available
20 firstTime = max(min(observations.Time), max(observations.Ti
21 recent = observations(observations.Time >= firstTime, :);
22
23 fprintf("Retrieved %d observations for station %s.\n", heig
24 fprintf("Displayed interval: %s to %s UTC.\n", ...
25     string(min(recent.Time)), string(max(recent.Time)))
26
27 %% Plot 1: wind speed and gusts
```

```
>> nbdc_chesapeake_bay_buoy_demo
Error using nbdc_chesapeake_bay_buoy_demo
The NDBC response contains incomplete
numeric observation rows.

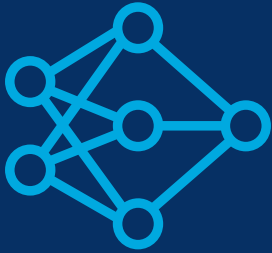
Error in nbdc_chesapeake_bay_buoy_demo (
observations = readNdbcRealtime(dataUrl)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
Explain Error
>>
```

Two Very Different Results

To this



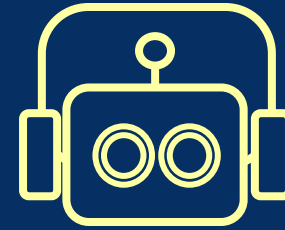
AI in Different *Flavors*



Discriminative AI
Map input to outputs to
make predictions



Generative AI
Generate content from
training data



Agentic AI
Plan and act using tools,
memory, and trained logic



?

Agents

Large language models

Machine learning + deep learning

Statistical Foundation

Modern AI is a stack, not a replacement sequence

What does a language model actually do?

“The vehicle speed is ...”



Next-token prediction = a probability distribution

A plausible answer, not necessarily the best/correct answer

“Why did the car crash?”



1. Failure to control speed
2. Driver inattention
3. Failure to yield correctly
4. Change lanes when not safe
5. Failure to drive in single lane

<https://desimonelawoffice.com/blog/top-5-contributing-factors-of-car-crashes-in-texas/>



← Evidence-based investigation →

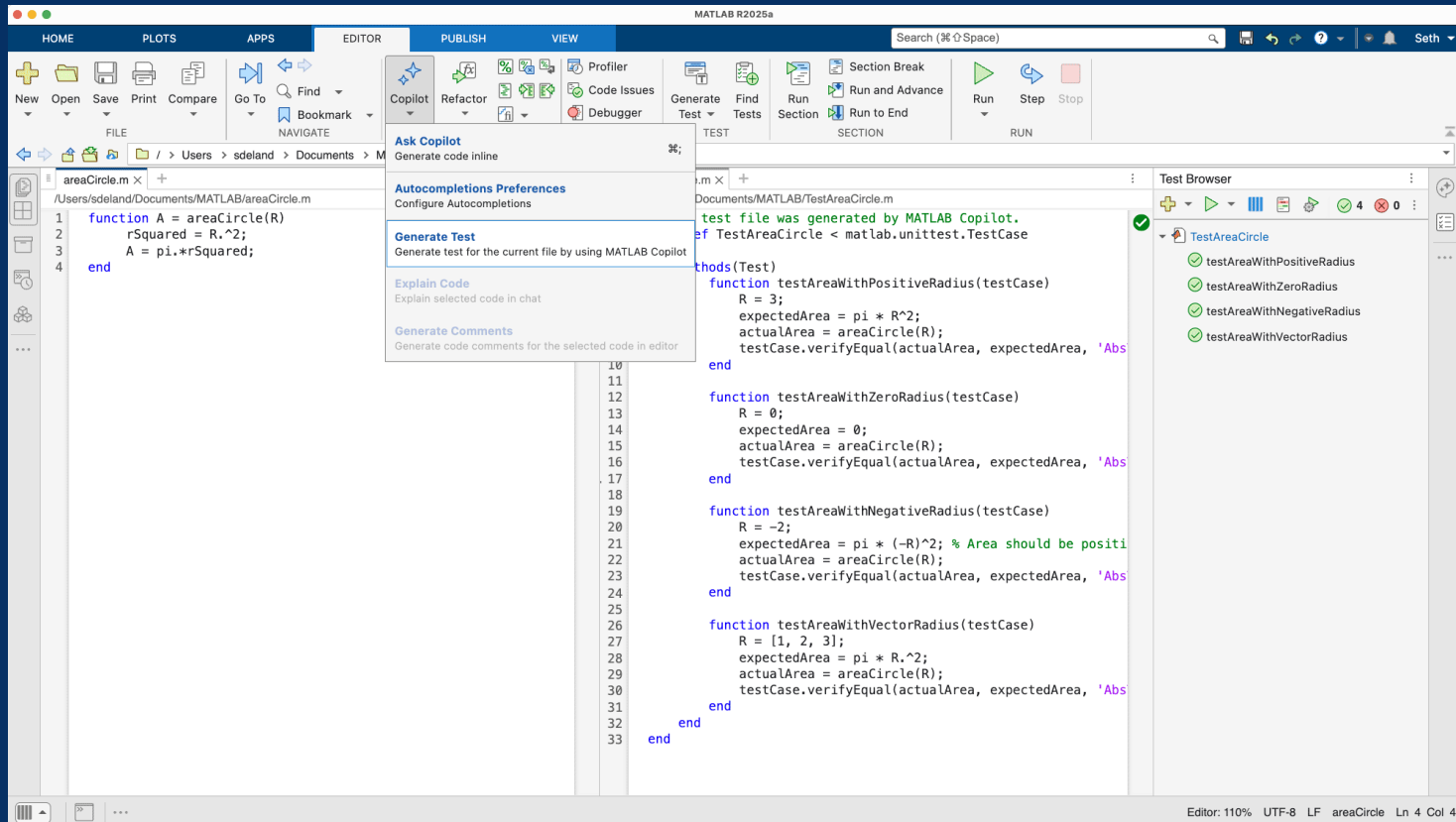


Did you inspect the model, logs, changes, and competing hypotheses?

Speed comes from the agent.
Trust comes from the verified workflow.
Intent and creativity come from you.



MATLAB Copilot



Copilot Chat



Copilot in the Editor
& Command Window



Copilot Actions



R2026a Simulink Copilot

Provides deep insights and actionable advice



Explain & Advise

- Learn tools & techniques
- Understand the model
- Search the model
- Get advice
- Troubleshoot errors

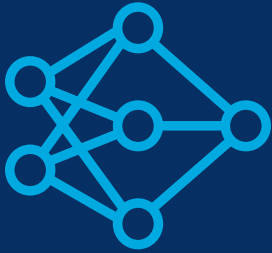


Review & Automate

- Automate tasks
- Track progress

The screenshot shows the Simulink environment with the 'BatteryManagementSystem' model open. The model diagram includes blocks for 'BMS_Algorithms', 'BMS_Info', 'BMS_Cmd', 'BatteryPlantModel', and 'KnownSOC'. The 'Copilot Chat' window is open on the right, displaying a welcome message from Simulink Copilot. The chat interface includes a text input field with the placeholder 'Ask Simulink Copilot', a 'Deeper Insights' button, and a 'Shuffle Example Prompts' button. The status bar at the bottom indicates 'Ready', '87%', and 'auto(ode3)'.

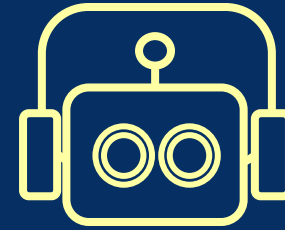
AI in Different *Flavors*



Discriminative AI
Map input to outputs to
make predictions



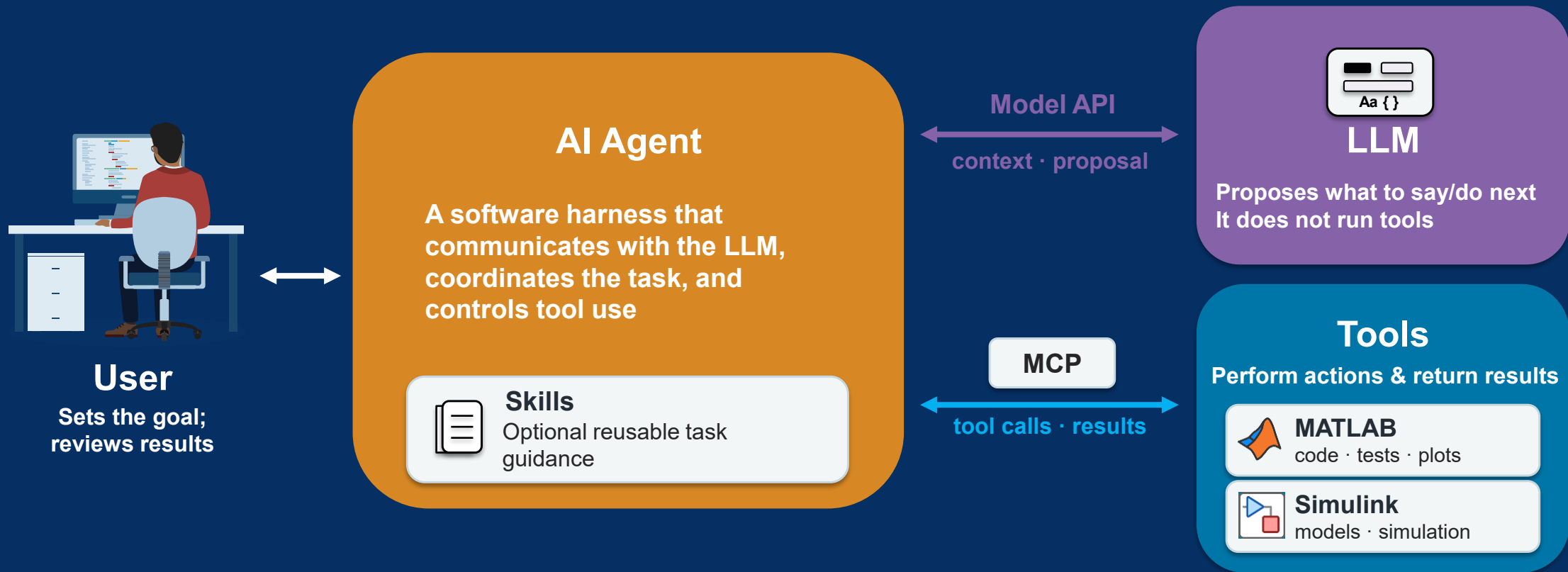
Generative AI
Generate new content
from training data



Agentic AI
Plan and act using tools,
memory, and trained logic

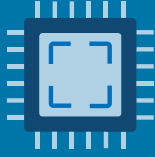


Agentic AI: the mental model



Agent coordinates · LLM proposes · Tools act · You review

M.E.R.L.I.N.



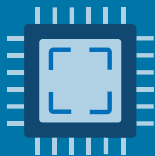
Modeling, Engineering, and Research Learning Intelligence Network

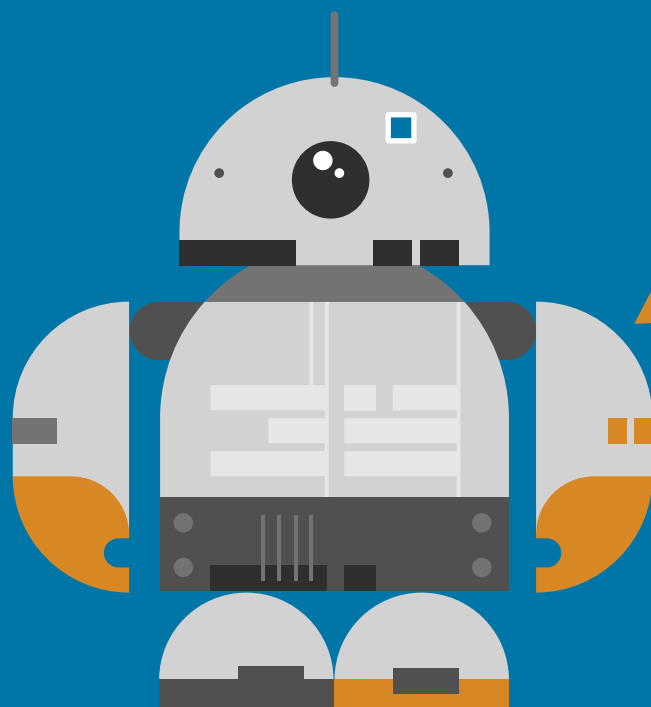
The LLM has

- Tons of knowledge examples
- Algorithms for parsing, reasoning, retrieving, evaluating
- Language processing

But it does not have

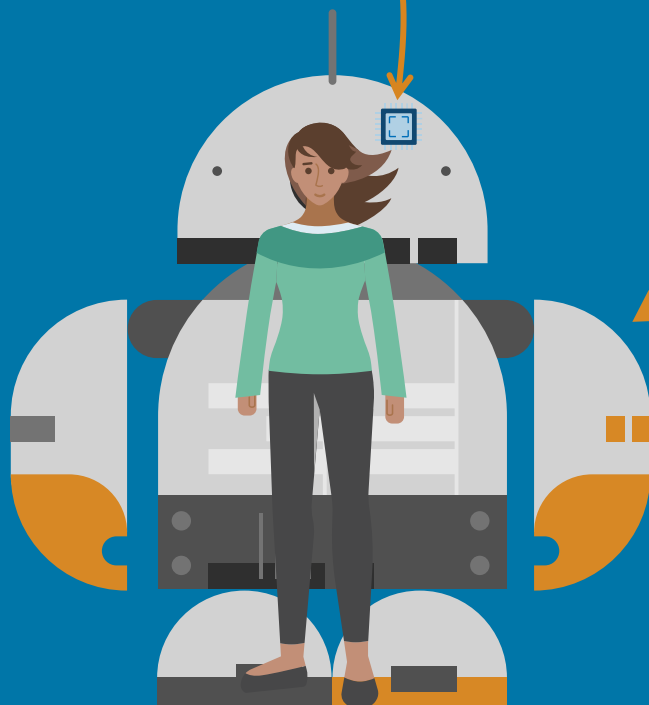
- Permission to access programs
- Ability to translate reasoning to action on computer/programs

M.E.R.L.I.N. 

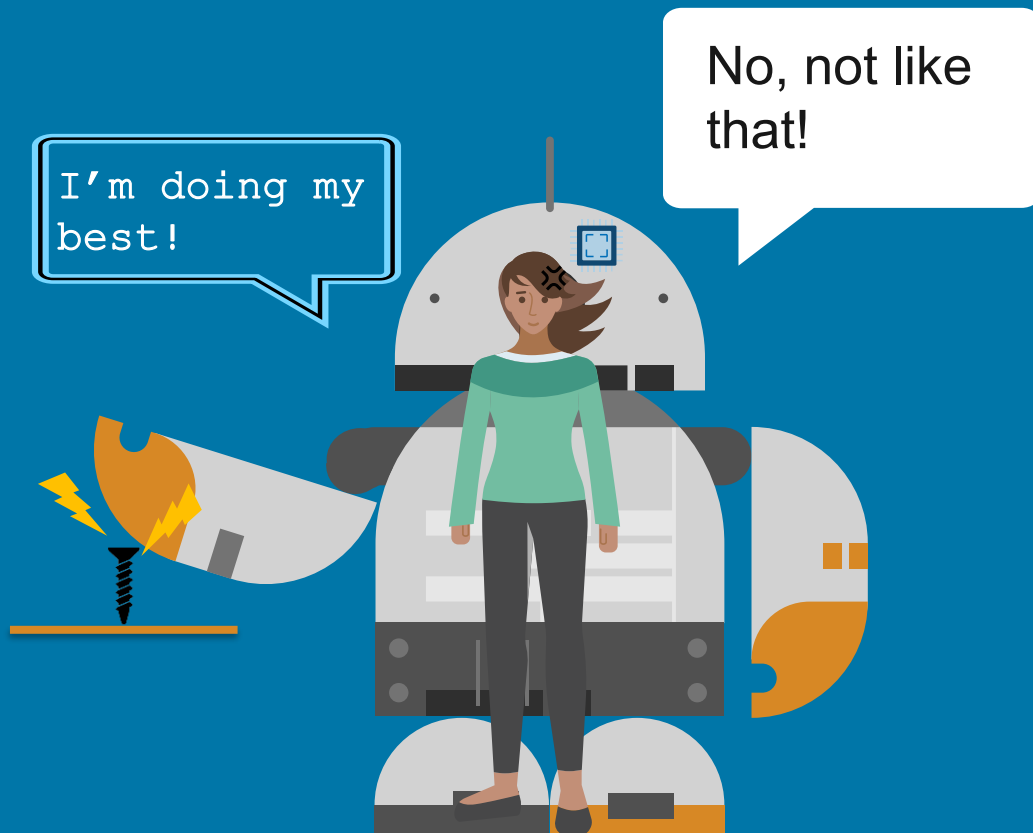


Science Suit Exoskeleton

M.E.R.L.I.N.



Science Suit Exoskeleton == the “agent” or “harness”

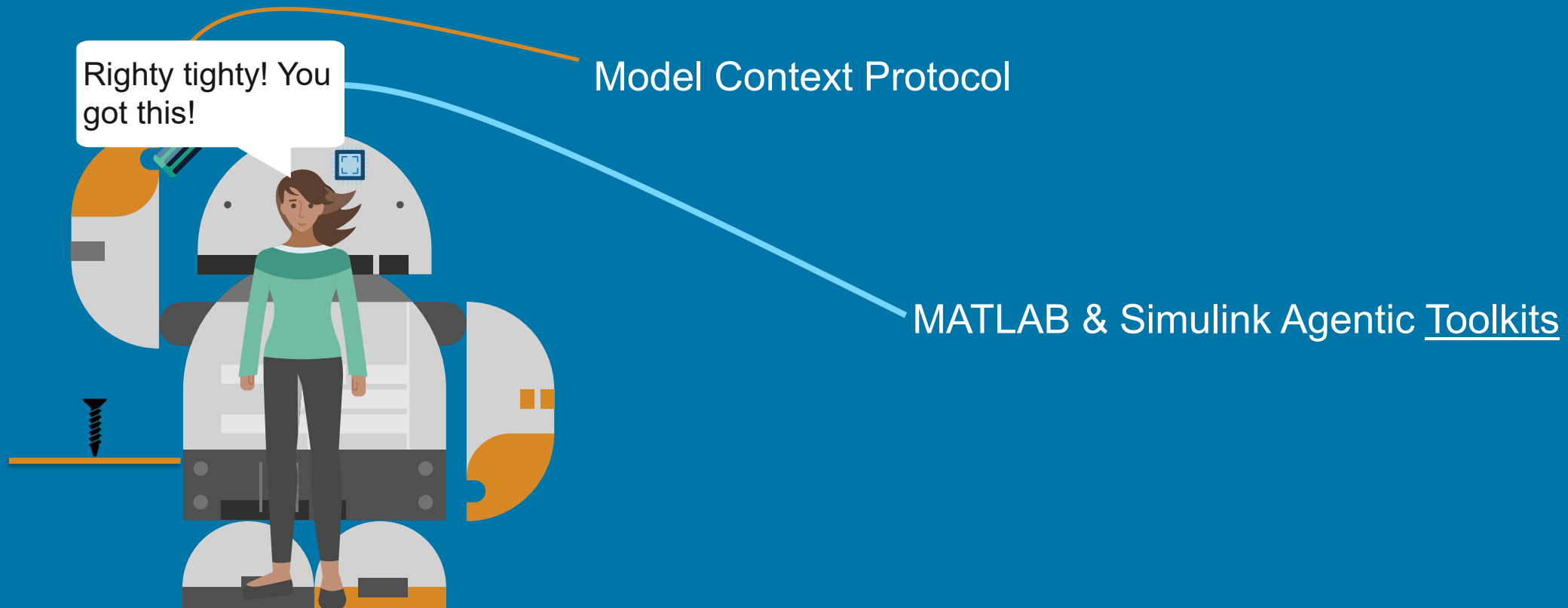


The agent/harness has

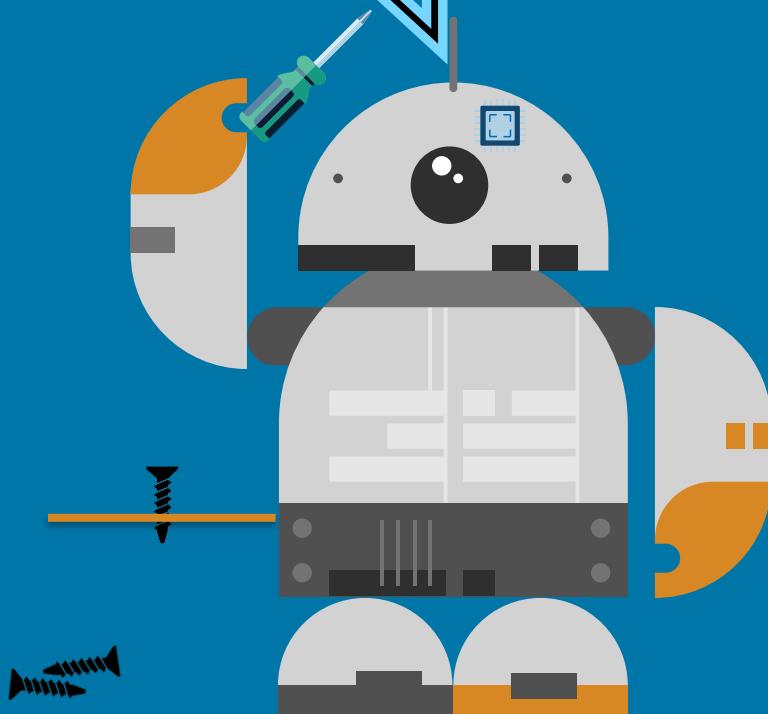
- A way to interact with your computer/command line

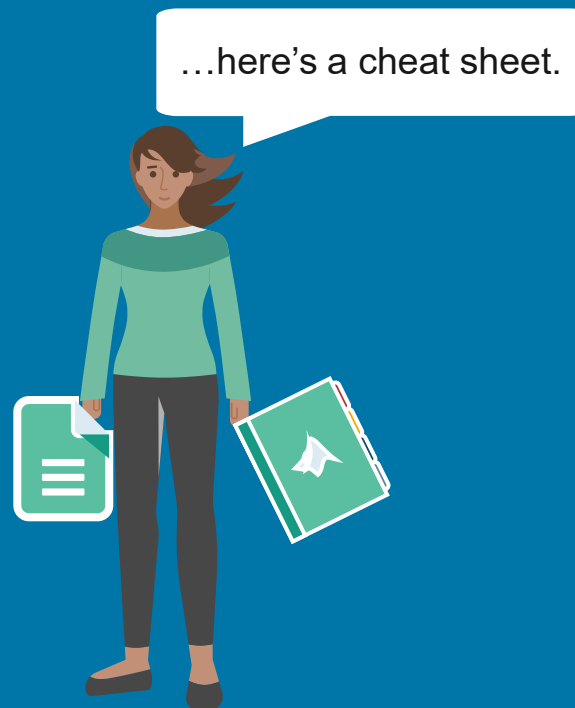
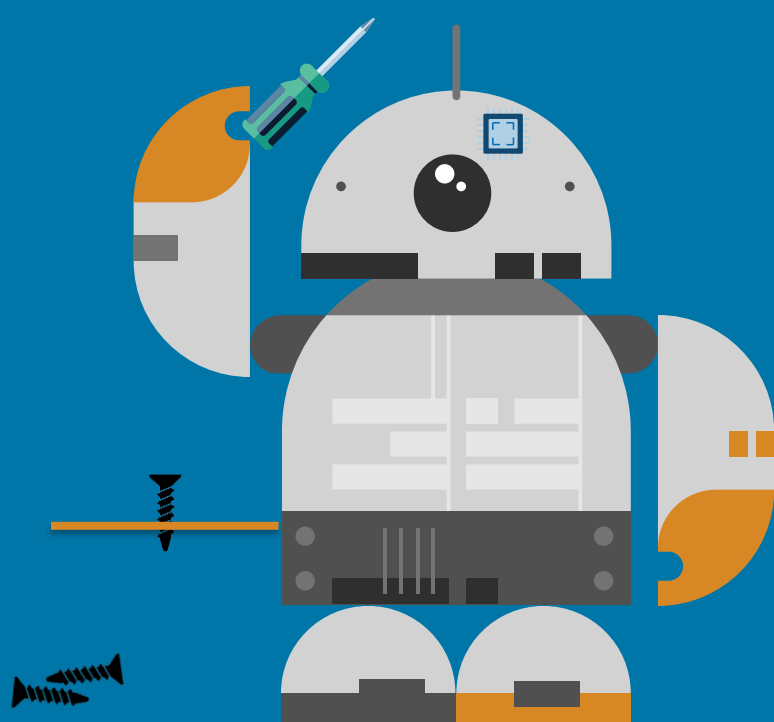
But it does not (necessarily) have

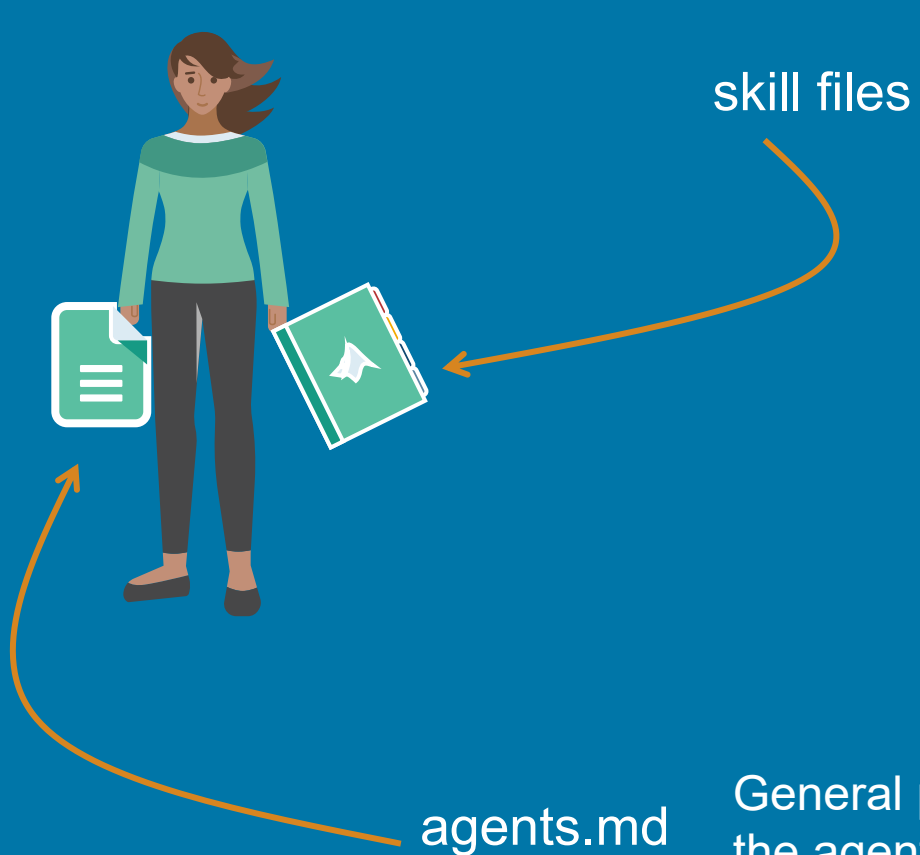
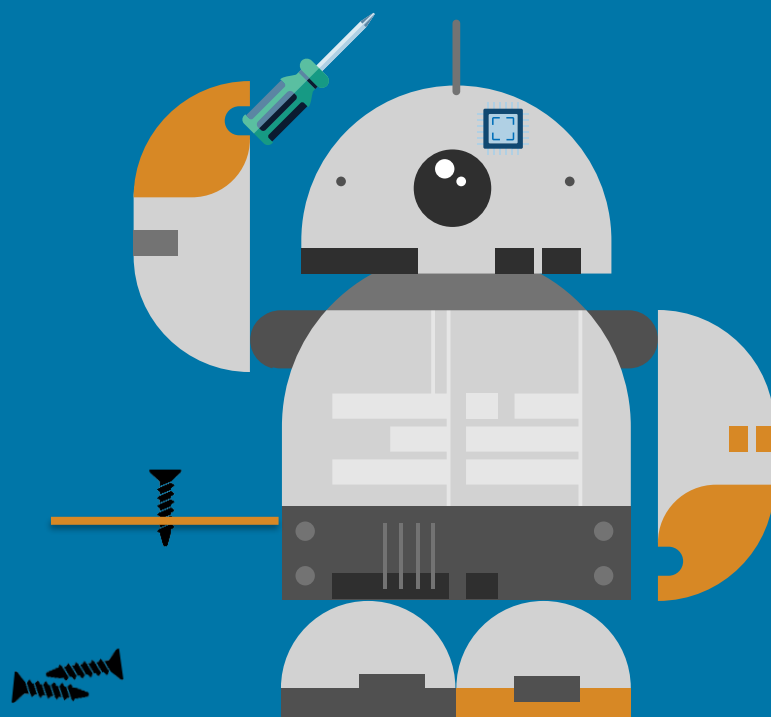
- Permission to access certain programs
- The correct tools for accomplishing tasks



I tried screwing it in both clockwise and counter-clockwise. It wouldn't go counterclockwise so, I increased the torque and stripped the screw. I wrote you an essay on all the steps I took to reach this point. I've got it now and am ready when you give me a new screw.

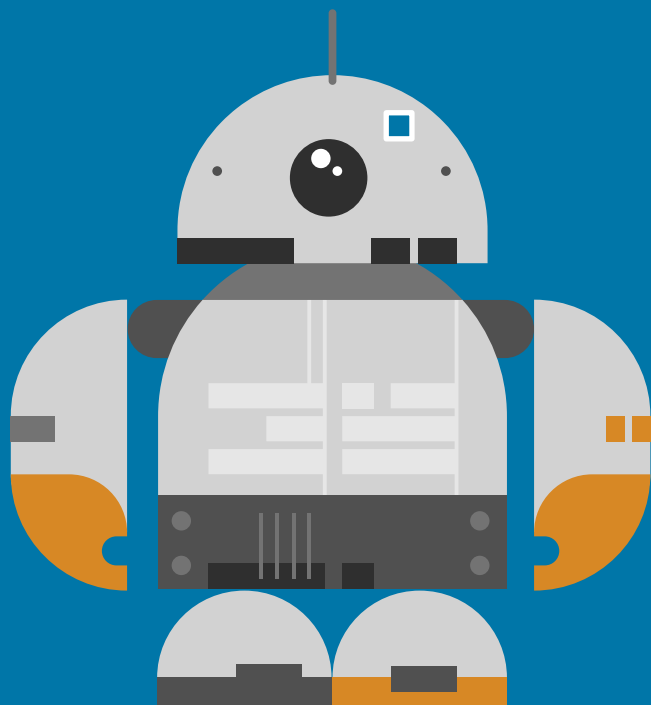




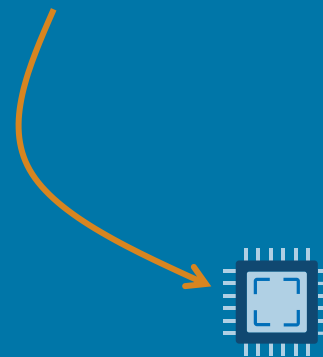


Best practices for
executing certain
tasks

General project guidance for
the agent

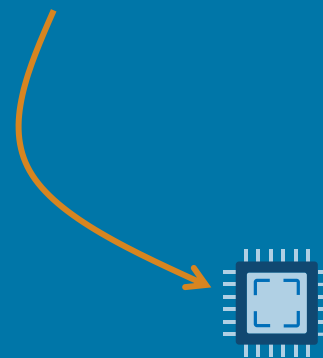
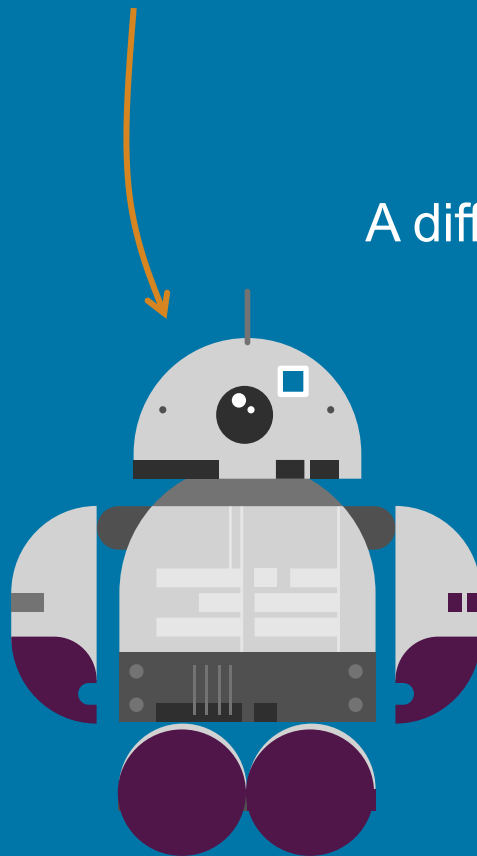
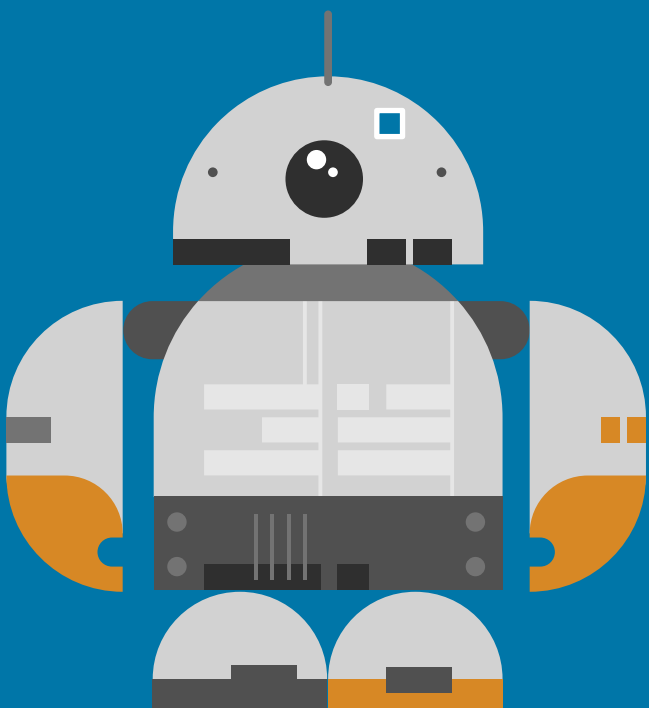


A different LLM

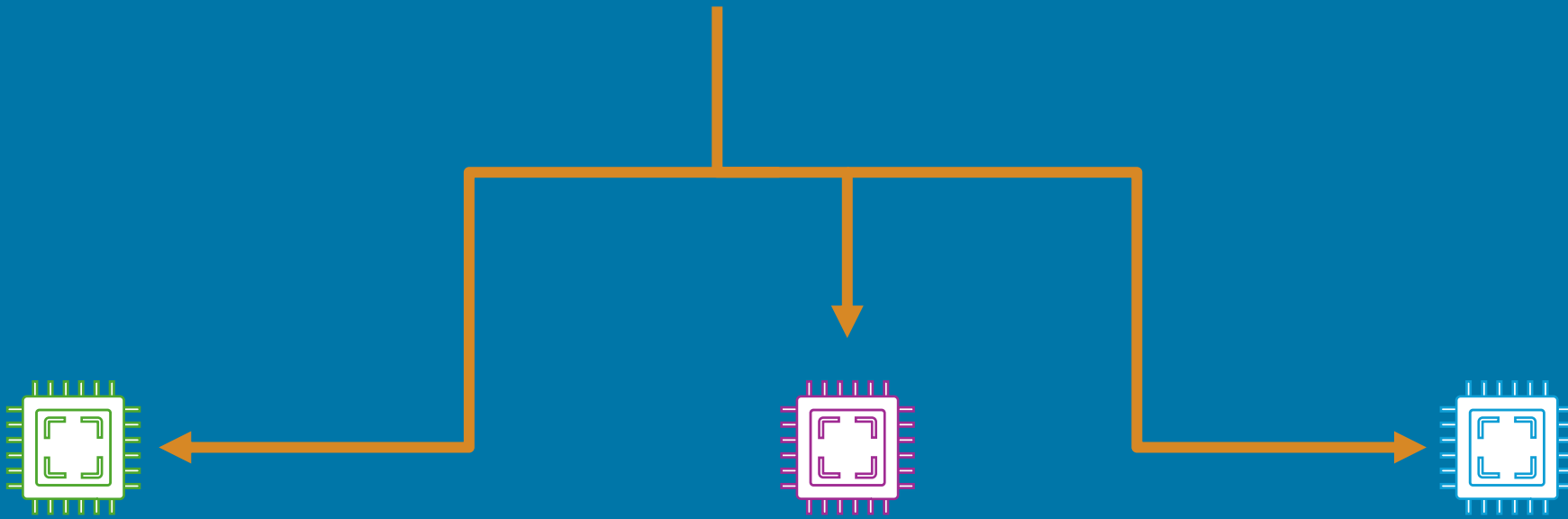
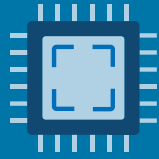


A different harness/agent

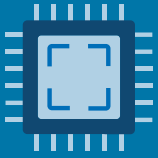
A different LLM



M.E.R.L.I.N.



Team orchestration



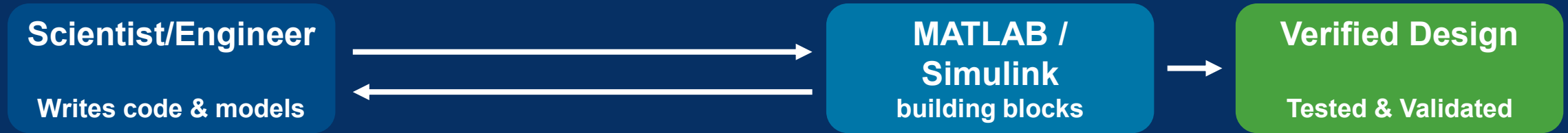
But who supervises the “Merlin”?

Speed comes from the agent.
Trust comes from the verified workflow.
Intent and creativity come from you.



AI changes creation speed, not accountability

Today

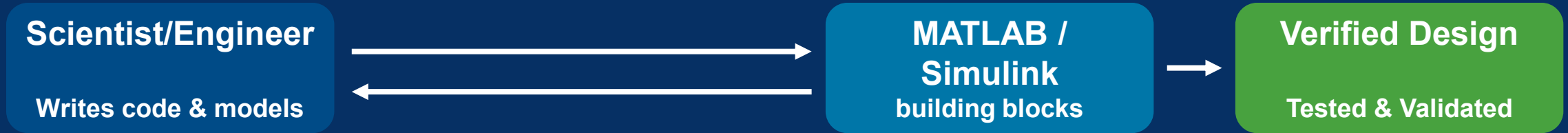


With AI



AI changes creation speed, not accountability

Today



With AI



What doesn't change when you add AI

You own the design. The same evidence verifies it.

Components to start using MATLAB with AI Coding Agents

1. An LLM that supports tool calling – e.g., **GPT 5.6**
2. AI Agent that supports MCP – e.g., **Codex CLI**
3. **Connection between MATLAB & AI Coding Agents via MCP**

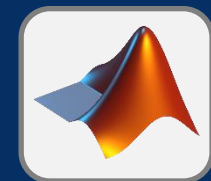
MATLAB Agentic Toolkit

Connect trusted MATLAB/Simulink capabilities to AI agents

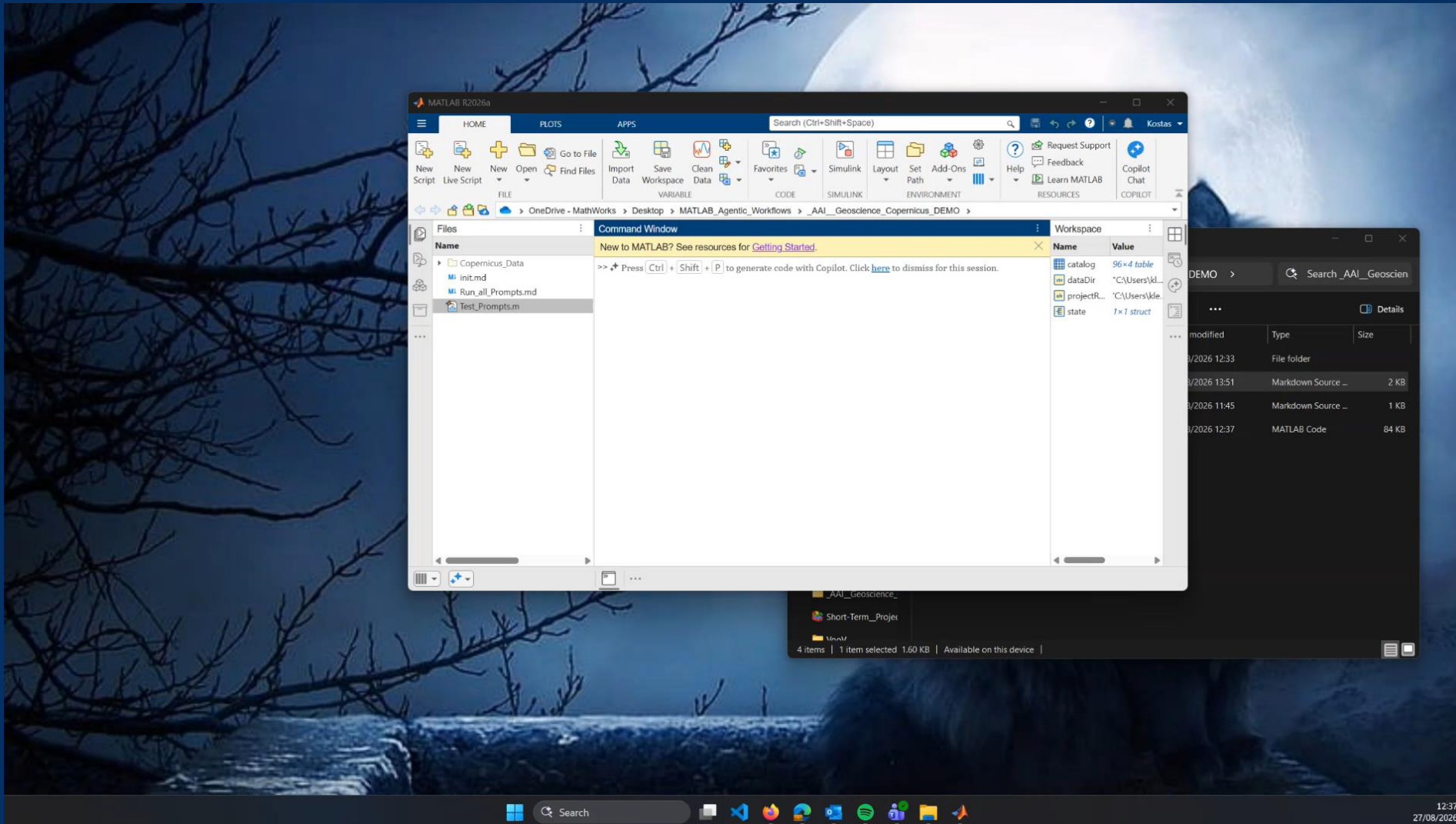
- A free, open-source local MCP server that connects GenAI assistants to a **locally licensed MATLAB**
- Expert Skills Catalog guides AI with best practices for testing, debugging, coding, and model creation.
- Repo + Installation instructions
<https://github.com/matlab/matlab-agentic-toolkit>



MATLAB Skills	
MATLAB MCP Server	
MATLAB MCP Tools	
• <code>evaluate_matlab_code</code> • <code>run_matlab_file</code> • <code>detect_matlab_toolboxes</code>	• <code>check_matlab_code</code> • <code>run_matlab_test_file</code>
Resources	
• <code>matlab_coding_guidelines</code>	• <code>plain_text_live_code_guidelines</code>
Simulink Skills	
Simulink MCP Tools	
• <code>model_overview</code> • <code>model_read</code> • <code>model_query_params</code> • <code>model_resolve_params</code>	• <code>model_edit</code> • <code>model_test</code> • <code>model_check</code>



Agentic AI in action – Kostas's Climate Example



Setup used

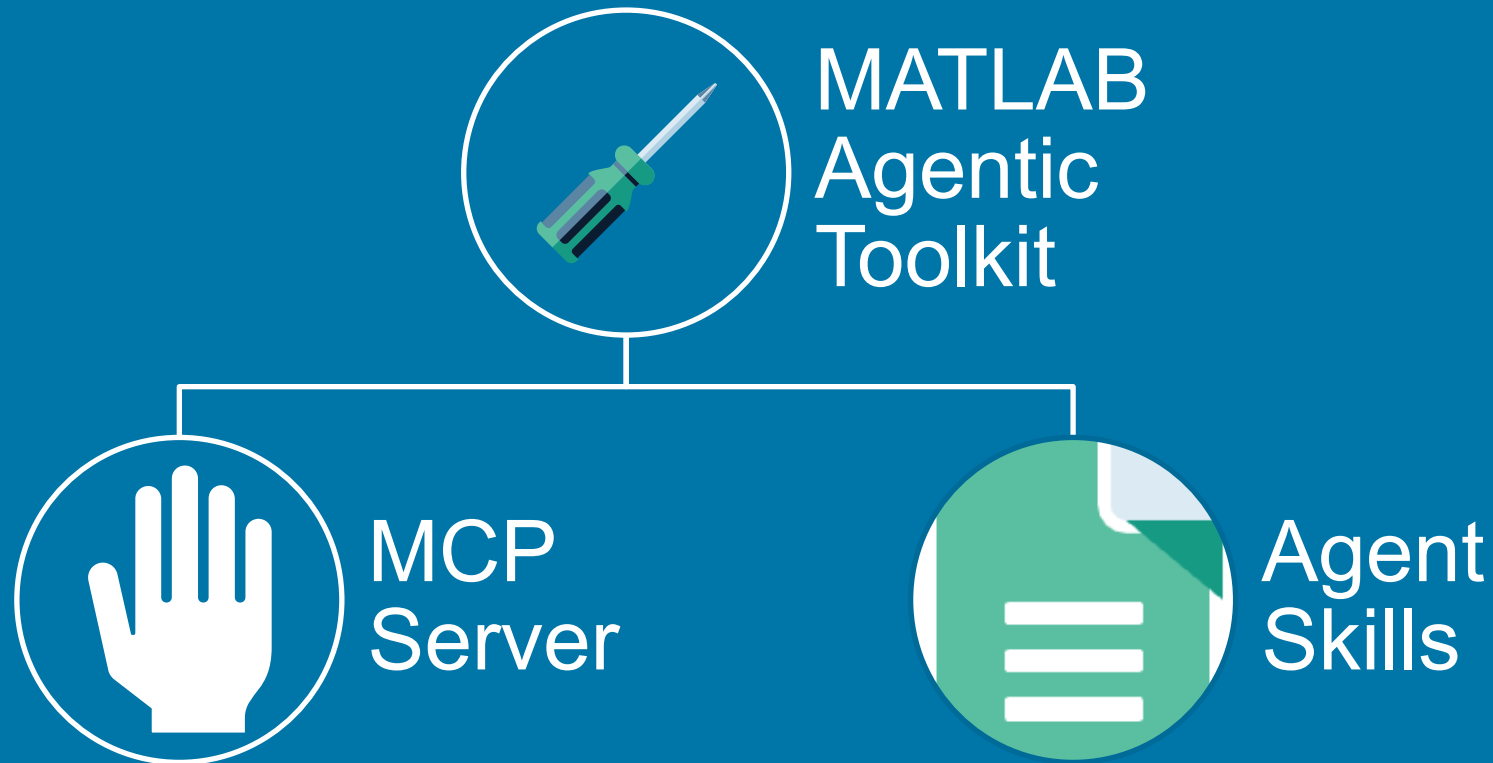
1. Agent: Codex
2. LLM: GPT 5.5
3. MATLAB R2026a
4. MATLAB Agentic Toolkit & Simulink Agentic Toolkit

What else can you do? **More like, what else can't you do...**

- Use other measured signals
 - Modify the physical model
 - Develop a new controller
 - Use a different model from our documentation
 - Use a wrapper to deploy app to mobile
 - Use MATLAB Mobile to capture sensor data
 - Add communication to local agent to interpret results
- [MATLAB Agentic Toolkit](#)
 - [Simulink Agentic Toolkit](#)
 - [MATLAB AI Agent SDK](#)
 - [Introduction: Agentic AI with MATLAB and Simulink](#)
 - [MathWorks Documentation](#)
 - [File Exchange](#)

Connect MATLAB w/AI Agents

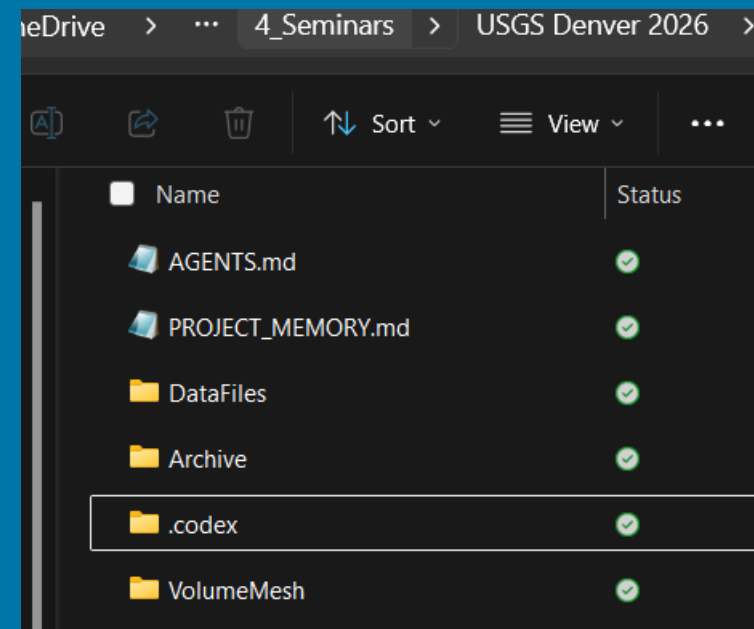
MATLAB Agentic Toolkit installation on GitHub



Your Agentic Environment

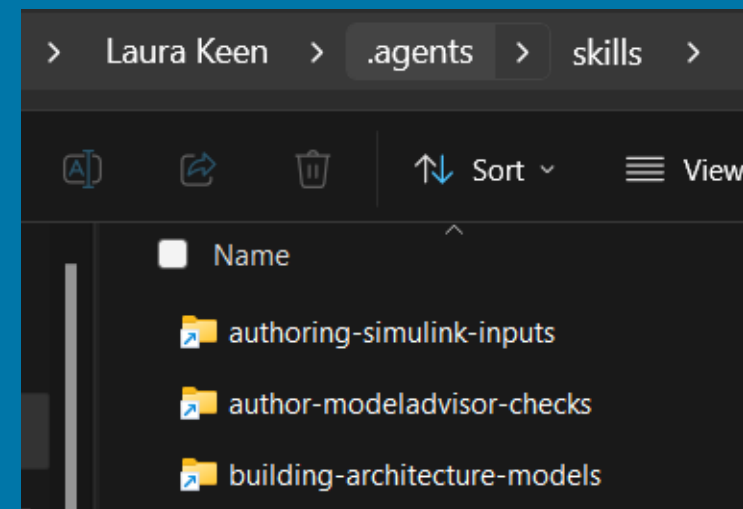
Working Directory

- Agents.md (v. specific)
- Skills.md (v. specific)
- Ref files (v. specific)



Global Directory

- Agents.md (global)
- Skills.md (global)
- Ref files (global)



Connect MATLAB w/AI Agents

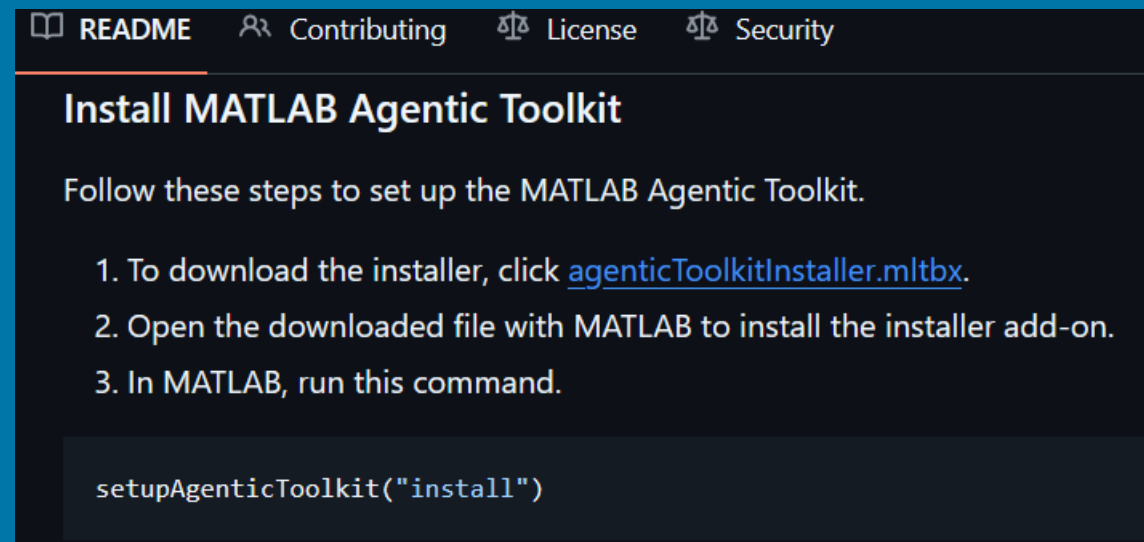
Option 1: Prompt your AI agent

- Clone/download the repo yourself & point your agent to the appropriate folder
 - <https://github.com/matlab/matlab-agentic-toolkit>
- "Setup/install the MATLAB Agentic Toolkit from this repository. Ask me for clarification if you have errors or questions."
 - "Follow the instructions for X."
 - "There are instructions for X. Adapt them for your use."
- Verify: "Is the MATLAB Agentic Toolkit properly installed? Can you access MATLAB and see the associated skill files? List the skills you can use."

Connect MATLAB w/AI Agents

Option 2: Manual Installation

- Download `agenticToolkitInstaller.mltbx`
 - Double-click the .mltbx file
 - MATLAB launches the installer
 - Accept the installation prompts
- Run in MATLAB command window:
 - `setupAgenticToolkit("install")`
- Select desired skill packs
- Verify by asking agent to interact with MATLAB



<https://github.com/matlab/matlab-agentic-toolkit/#install-matlab-agentic-toolkit>

Installer configures
everything automatically

Sharing MATLAB Session w/Your Agent

If MATLAB is already open, who gets priority?

- Agent cannot see a MATLAB session that doesn't have MATK running already
 - >matk_initialize
- Have matk_initialize in your startup.m file
- Put setup and sharing instructions in the agents.md file or related project files

```
% Initialize Simulink Agentic Toolkit (which also makes sure MATLAB Agentic  
% Toolkit is enabled)  
% 1. Initialize the Simulink Agentic Toolkit:  
%addpath("C:\Users\lsammon\.matlab\agentic-toolkits\simulink")  
%satk_initialize
```

- Or manually share session
 - MATLAB command line:
 > matlab.engine.shareEngine()

Efficient Agentic Workflows for Science & MATLAB

- Project setup, scope, and source control
- What should your agent do each time you launch a session?

No single industry-standard format yet!

but

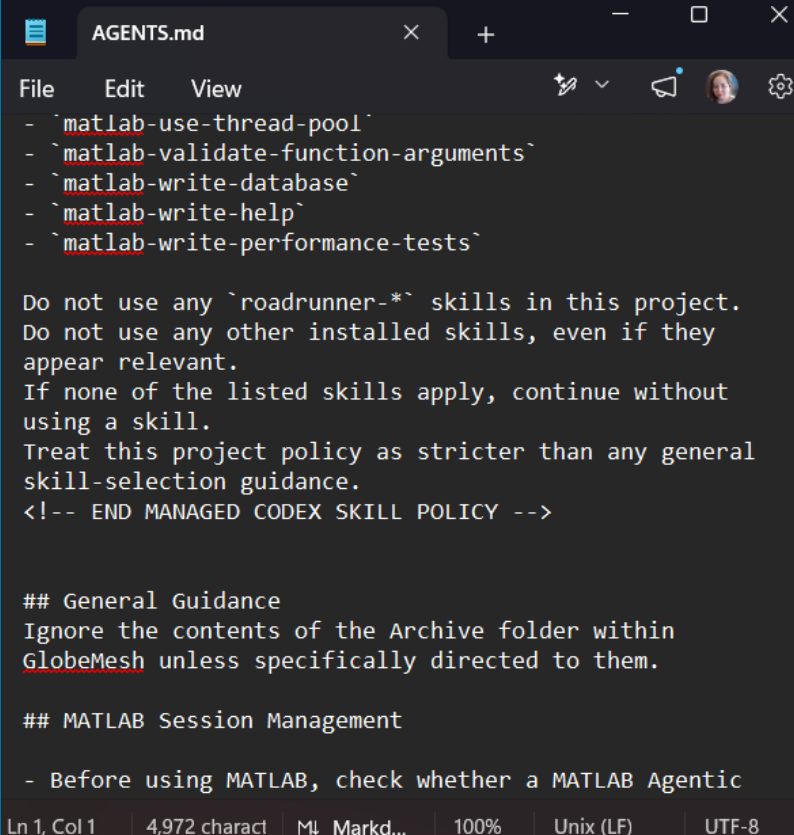
rapidly converging around SKILL.md and AGENT.md files with clear triggers, workflows, examples, and validation steps.

Efficient Agentic Workflows for Science & MATLAB

Recommend keeping project work in a dedicated workspace, for example:

- ~/Projects/
 - MyProject/
 - AGENTS.md
 - code/
 - data/
 - otherFolders/

Each project gets its own: AGENTS.md that defines project-specific behavior.



```
AGENTS.md
File Edit View
- `matlab-use-thread-pool`
- `matlab-validate-function-arguments`
- `matlab-write-database`
- `matlab-write-help`
- `matlab-write-performance-tests`

Do not use any `roadrunner-*` skills in this project.
Do not use any other installed skills, even if they
appear relevant.
If none of the listed skills apply, continue without
using a skill.
Treat this project policy as stricter than any general
skill-selection guidance.
<!-- END MANAGED CODEX SKILL POLICY -->

## General Guidance
Ignore the contents of the Archive folder within
GlobeMesh unless specifically directed to them.

## MATLAB Session Management

- Before using MATLAB, check whether a MATLAB Agentic
```

Ln 1, Col 1 | 4,972 charact | M Markd... | 100% | Unix (LF) | UTF-8

Efficient Agentic Workflows for Science & MATLAB

- Frame problems like an engineer for agentic workflows
 - Project requirements, specifications
 - Scope
 - Milestones
- Evaluate, test, and document agent-generated code
 - What's a unit test?
 - Programming best practices

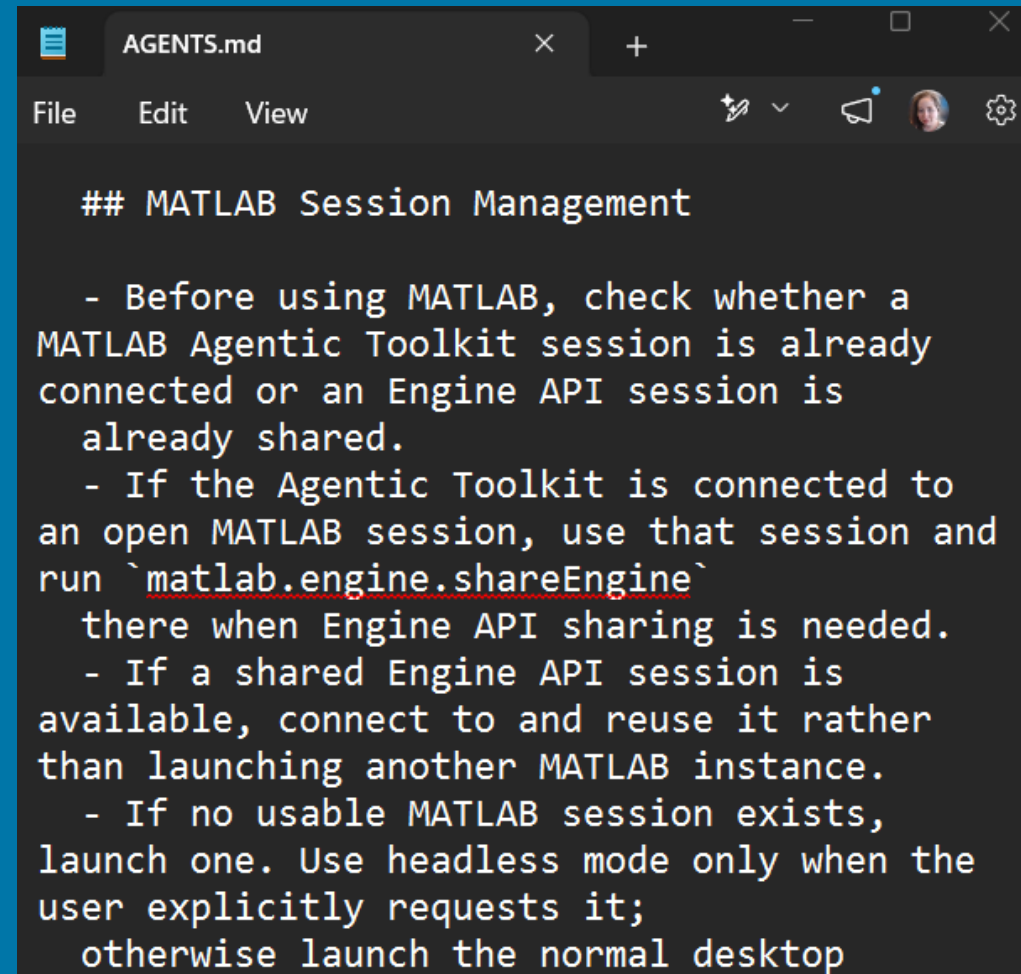


Skills, Context, Customization

- How to recognize common tasks
- Process to follow & preferred standards
- Create domain-specific workflows
- Automate quality checks
- Establish document templates
- Capture organizational knowledge

Setting Up for Success – Agents.md

- Keep it short
- Project Overview
- Working Style
- Quality Expectations
- Validation
- Deliverables
- Project-Specific Guidance



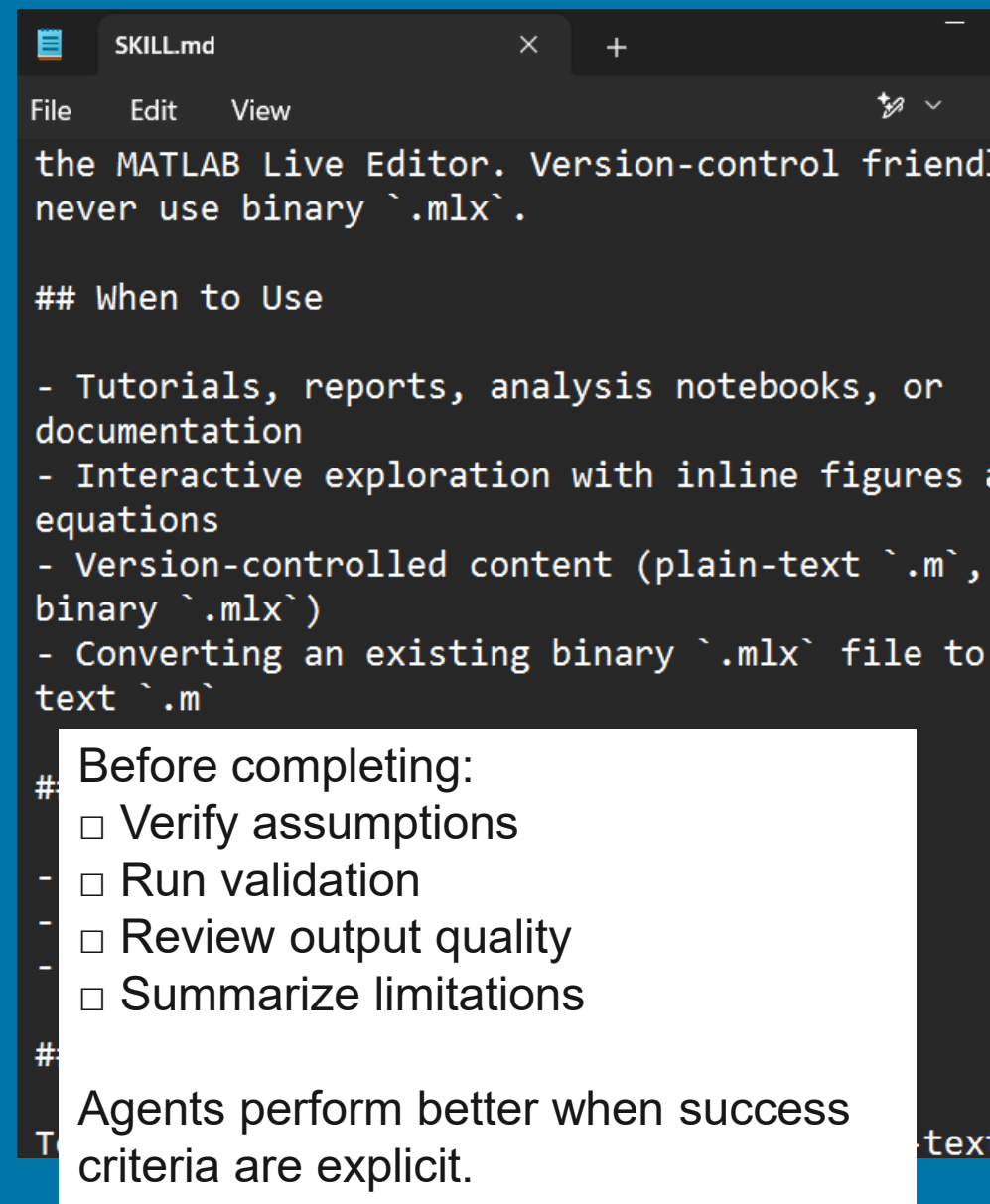
The screenshot shows a MATLAB code editor window titled 'AGENTS.md'. The editor contains the following text:

```
## MATLAB Session Management
```

- Before using MATLAB, check whether a MATLAB Agentic Toolkit session is already connected or an Engine API session is already shared.
- If the Agentic Toolkit is connected to an open MATLAB session, use that session and run ``matlab.engine.shareEngine`` there when Engine API sharing is needed.
- If a shared Engine API session is available, connect to and reuse it rather than launching another MATLAB instance.
- If no usable MATLAB session exists, launch one. Use headless mode only when the user explicitly requests it; otherwise launch the normal desktop

Setting Up for Success – Skills.md

- Overview
 - What problem does this solve?
- When to Use & Not Use
 - Specific trigger conditions
 - Avoid incorrect invocation
- Workflow
- Output Requirements
 - Required deliverables and format
- Examples
- Validation Checklist
 - Quality gates before completion



```
SKILL.md
File Edit View
the MATLAB Live Editor. Version-control friendly
never use binary `.mlx`.

## When to Use

- Tutorials, reports, analysis notebooks, or
documentation
- Interactive exploration with inline figures &
equations
- Version-controlled content (plain-text `.m`,
binary `.mlx`)
- Converting an existing binary `.mlx` file to
text `.m`

# Before completing:
- ☐ Verify assumptions
- ☐ Run validation
- ☐ Review output quality
- ☐ Summarize limitations

# Agents perform better when success
criteria are explicit.
```

Resources & Links

[MATLAB Agentic Toolkit installation on GitHub](#)

<https://www.mathworks.com/campaigns/offers/next/agentic-ai-with-matlab.html>

<https://claude.com/blog/steering-claude-code-skills-hooks-rules-subagents-and-more>

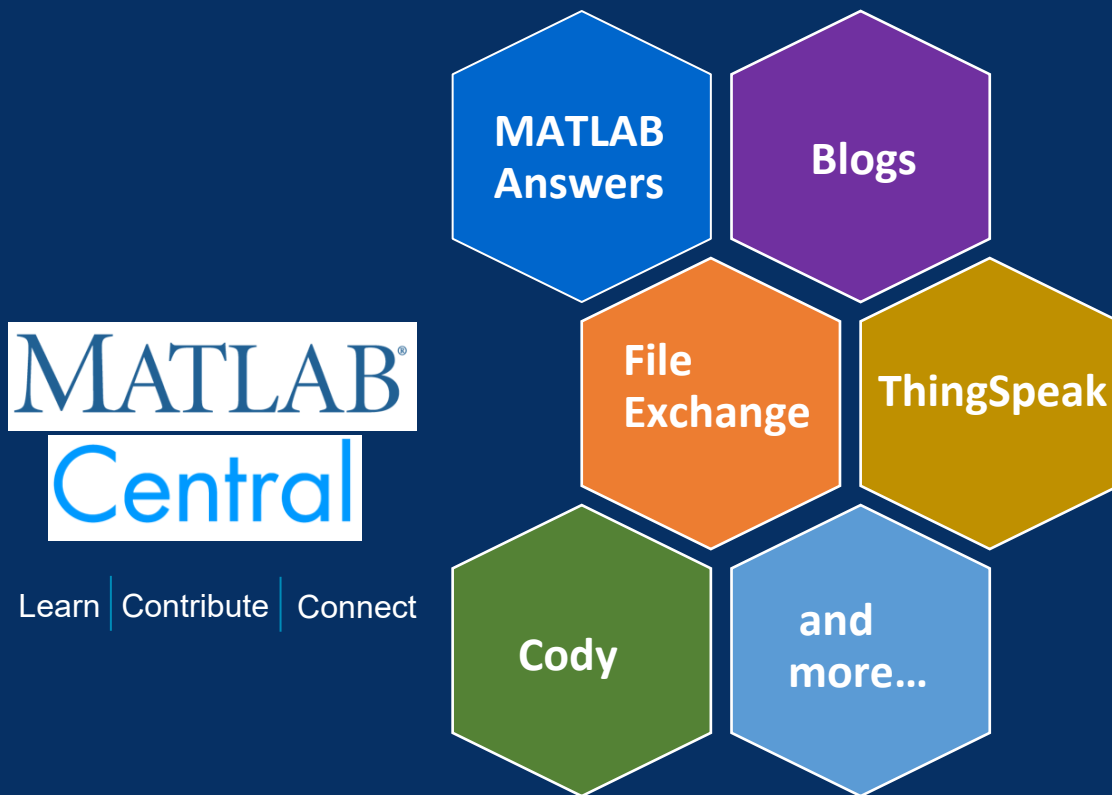
<https://learn.chatgpt.com/guides/best-practices>

<https://github.com/openai/skills>

<https://code.claude.com/docs/en/skills>

MATLAB Central Community

Every month, over **2 million** MATLAB & Simulink users visit MATLAB Central to get questions answered, download code and improve programming skills.



MATLAB Answers: Q&A forum; most questions answered in < **1 hour**

File Exchange: Download code from a huge repository of free code including **tens of thousands** of open source community files

Cody: Sharpen programming skills while having fun

Blogs: Get the inside view from Engineers who build and support MATLAB & Simulink

ThingSpeak: Explore LoT Data



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